Stieltjes moment classification of Apéry polynomials

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Abstract

We prove that the Apéry polynomials $A_n(x) = \sum_{k=0}^{n} \binom{n}{k}^2 \binom{n+k}{k}^2 x^k$ form a Stieltjes moment sequence in n, for real $x \geq 0$, exactly when $x = 0$ or $x \geq 1$. This settles Sokal's conjecture for real parameters; the case $x = 1$ is Edgar's theorem on the Apéry numbers. Both directions start from a single representation of $A_n(x)$ as the Legendre moments of an explicit probability law on two angles. For $0 < x < 1$, a Stieltjes representation would force the part of this law below 1 to have a real-analytic density. The law instead has a support gap when $x < 1/4$, a saddle singularity when $1/4 < x < 1$, and a degenerate endpoint at $x = 1/4$, so no such representation exists. For $x > 1$, the same part of the law has a density given by a complete elliptic integral. We continue it to the plane cut along $[1, \infty)$, show that its jump across the cut is positive, and recover it by Cauchy's formula, using a uniform bound on large circles. Heine's integral for the Legendre functions Q_n then turns the remaining Legendre moments into the power moments of a positive measure on $[0, 1]$. We also give exact rational certificates showing that, for each $0 < x \leq 2/3$, one of four explicit Hankel minors is negative.

Keywords: Apéry polynomials; Stieltjes moment problem; Legendre polynomials; elliptic integrals; Hankel matrices.

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1 Introduction

A real sequence $(a_n)_{n\geq 0}$ is a *Stieltjes moment sequence* if $a_n = \int_0^\infty t^n \, d\nu(t)$ for a positive Borel measure ν. Define

$$A_n(x) = \sum_{k=0}^{n} \binom{n}{k}^2 \binom{n+k}{k}^2 x^k. \tag{1}$$

Wang and Zhu posed moment questions for generalized Apéry polynomials, including the family (1) [9, Section 5]. Sokal recorded the case $x \geq 1$ as an open problem [7] and later conjectured that $(A_n(x))_{n\geq 0}$ is a Stieltjes moment sequence for every $x \geq 1$ and is not one for $0 < x < 1$ [8, slide 29]. Edgar proved the case $x = 1$ [3]. Our main result is the full classification for real parameters.

Theorem 1. *For real $x \geq 0$, $(A_n(x))_{n\geq 0}$ is a Stieltjes moment sequence if and only if $x = 0$ or $x \geq 1$.*

Sokal's stronger conjecture, that the Hankel matrix of $(A_n(1 + x))_{n\geq 0}$ is coefficientwise totally positive as a matrix of polynomials in x, remains open.

1.1 Outline of the proof

Throughout, P_n is the Legendre polynomial and $r = \sqrt{x}$. Both ranges of x are handled through the identity $A_n(x) = \int P_n(y)\, d\mu_r(y)$ of Section 3, where μ_r is an explicit probability measure with compact support. The part of μ_r on $[1, \infty)$ already gives positive power moments, by Laplace's integral for P_n. Everything therefore depends on the part on $[-1, 1)$.

For $0 < r < 1$, the Poisson kernel for Legendre series shows that a Stieltjes representation would force this part to have a real-analytic density (Section 2). The geometry of μ_r rules this out: its support has a gap when $r < 1/2$, it has a saddle singularity when $r > 1/2$, and it has a quartic endpoint when $r = 1/2$ (Section 4). For $r > 1$, the density is a complete elliptic integral. We continue it to the plane cut along $[1, \infty)$, compute its jump across the cut, which is positive, and recover the density by Cauchy's formula. The classical Q_n transform then turns the remaining Legendre moments into power moments of a positive measure on $[0, 1]$ (Section 5). Section 6 gives exact Hankel certificates for $0 < x \le 2/3$ and explains why no fixed finite test can cover the whole interval $(0, 1)$.

1.2 Related work

Common showed that Legendre series with Stieltjes coefficients continue to a cut plane, and that positive mixtures of the functions Q_n give Stieltjes moment sequences [1, CERN preprint, Theorems 2–3 and 5]; he notes that Yndurain found the second result independently. We give the finite-measure and endpoint details needed in our setting. The new ingredients are the geometric obstruction for $0 < x < 1$ and the elliptic boundary calculation for $x > 1$.

At $x = 1$, our angular representation reduces to a formula attributed to Lupaş (2005) and proved by Ivanisvili in 2014 [2]; the product and generating-function identities behind it are classical. Matala-aho derived a recurrence for the same polynomials [4, Theorem 2], and Neuschel studied their asymptotics and the distribution of their zeros [5]; we do not need either result. Zhu and Sun studied arithmetic properties of the Hankel determinants of the Apéry numbers [10].

2 A regularity obstruction

The Legendre polynomials are normalized by $P_n(1) = 1$. We first record the consequence of the Poisson kernel that we need.

Lemma 2. *Let μ be a finite positive measure supported on $[-1, H]$, where $H \ge 1$. If $a_n = \int P_n(y)\, d\mu(y)$ is a Stieltjes moment sequence, then $\mu|_{[-1,1)}$ has a density real analytic on $(-1, 1)$ and bounded in a right neighborhood of -1.*

Proof. For $y \ge 1$, Laplace's integral representation gives

$$P_n(y) = \frac{1}{\pi} \int_0^\pi \left(y + \sqrt{y^2 - 1}\cos\phi\right)^n d\phi. \tag{2}$$

Hence the contribution from $[1, H]$ is the moment sequence of a finite positive measure ν_+ on $[0, R]$, where we may take $R = \max\{2, H + \sqrt{H^2 - 1}\}$. Since $|P_n(y)| \le 1$ on $[-1, 1]$ and $|P_n(y)| \le R^n$ on $[1, H]$, we have $|a_n| \le \mu([-1, H])R^n$. Any positive Stieltjes measure ν representing (a_n) must therefore be supported on $[0, R]$: positive mass beyond $R + \delta$, for any $\delta > 0$, would contradict this exponential bound.

The finite signed measure $\sigma = \nu - \nu_+$ has moments

$$b_n = \int_{[-1,1)} P_n(y)\, d\mu(y), \qquad |b_n| \le \mu([-1,1)). \tag{3}$$

These bounds force σ to be supported on $[0,1]$, even though σ need not be positive. Indeed, its Cauchy transform agrees, for $|z| > R$, with $\sum_{n\ge 0} b_n z^{-n-1}$. The series is holomorphic on $|z| > 1$, so by the identity theorem the two functions agree on $\{\operatorname{Im} z > 0,\ |z| > 1\}$, and likewise in the lower half-plane. The series is real on $(1,\infty)$. The Stieltjes inversion formula, which holds for finite signed measures by dominated convergence against $|\sigma|$, therefore gives $\sigma(I) = 0$ for every interval $I \subset (1,\infty)$ whose endpoints are not atoms of σ. Such intervals exhaust $(1,\infty)$, so σ vanishes there.

For $0 \le s < 1$ put

$$K(s,y) = \frac{1-s^2}{2(1-2sy+s^2)^{3/2}}, \qquad -1 \le y \le 1. \tag{4}$$

The generating function for P_n, followed by differentiation and orthogonality, gives

$$K(s,y) = \frac{1}{2}\sum_{n\ge 0}(2n+1)s^n P_n(y), \qquad \int_{-1}^{1} P_n(y)K(s,y)\, dy = s^n.$$

In particular, $K(s,y)\, dy$ is a probability measure. For $s = 1$ use δ_1 instead. Integrating these measures against σ produces a signed measure η of total variation at most $|\sigma|([0,1])$. Fubini's theorem applies because $|P_n| \le 1$ on $[-1,1]$, and shows that the Legendre moments of η are the b_n. Since the Legendre polynomials span all polynomials, uniqueness of finite signed measures on this compact interval implies $\eta = \mu|_{[-1,1)}$.

Since $\mu|_{[-1,1)}$ has no mass at 1, the atom $\sigma(\{1\})\delta_1$ vanishes, and $\mu|_{[-1,1)}$ has density $\int_{[0,1)} K(s,y)\, d\sigma(s)$. For each real $y_0 < 1$, $1 - 2sy_0 + s^2$ has a positive minimum over $0 \le s \le 1$. Thus the kernels are holomorphic and uniformly bounded in a small complex neighborhood of y_0. Integration against the finite signed measure preserves holomorphy. This proves analyticity on $(-1,1)$ and, by taking $y_0 = -1$, boundedness near the left endpoint. □

3 The angular representation

The classical identities used in this section follow from the Legendre addition theorem and integral representations; see [6]. The finite expansion

$$P_n(1+2z) = \sum_{k=0}^{n} \binom{n}{k}\binom{n+k}{k} z^k$$

and Parseval's identity imply, with $r = \sqrt{x}$,

$$A_n(x) = \mathbb{E}_t \left| P_n(1 + 2re^{it}) \right|^2, \tag{5}$$

where t is uniform on $[0, 2\pi]$. The Legendre product formula is

$$P_n(a)P_n(b) = \frac{1}{\pi}\int_0^{\pi} P_n\left(ab + \sqrt{a^2-1}\sqrt{b^2-1}\cos\theta\right) d\theta. \tag{6}$$

Although initially stated for real arguments, this is a polynomial identity in a, b: after integration, only even powers of the product of square roots remain. It therefore extends to complex arguments

and is independent of the choices of square roots. For $b = \overline{a}$, choose conjugate square roots. Equations (5)–(6) yield

$$A_n(x) = \mathbb{E}P_n(Y_r), \tag{7}$$

where t, θ are independent and uniform on $[0, 2\pi]$, and

$$Y_r = 1 + 4r^2 + 4r\cos t + 4r\sqrt{1 + 2r\cos t + r^2}\cos\theta. \tag{8}$$

(Taking θ uniform on $[0, 2\pi]$ rather than $[0, \pi]$ does not change the distribution of $\cos\theta$.) Writing $a = 1 + 2re^{it}$, we have $Y_r = |a|^2 + |a^2 - 1|\cos\theta$. Since $|a^2 - 1| \le |a|^2 + 1$, $|a| \le 1 + 2r$ and $|a^2 - 1| = 4r|1 + re^{it}| \le 4r(1 + r)$, we have

$$-1 \le Y_r \le 1 + 8r^2 + 8r. \tag{9}$$

For $0 < r < 1$, the square root in (8) is strictly positive and smooth.

When $r = 1$, substituting $t = 2\alpha$ in (8) gives the same distribution as $1 + 8\cos^2\alpha + 8\cos\alpha\sin\beta$ with independent uniform angles: the absolute value of $\cos\alpha$ is absorbed by symmetry in the second angle. This is the formula of Lupaş [2]; the proof does not use it.

4 Failure on $0 < x < 1$

Theorem 3. *For every fixed real x with $0 < x < 1$, the sequence $(A_n(x))_{n\ge0}$ is not a Stieltjes moment sequence.*

Proof of Theorem 3. Fix $0 < r < 1$, and suppose that $(A_n(x))_{n\ge0}$ is a Stieltjes moment sequence. Lemma 2 applies to the distribution μ_r of Y_r. Set

$$h = 1 + 8r^2 - 8r.$$

At $(t, \theta) = (\pi, \pi)$, $Y_r = h < 1$, so by continuity the law has positive mass below 1. We distinguish three cases.

Case 1: $0 < r < 1/2$**.** For $c = \cos t$, the lower endpoint as θ varies is

$$f(c) = 1 + 4r^2 + 4rc - 4r\sqrt{1 + 2rc + r^2}.$$

Since

$$f'(c) = 4r\left(1 - \frac{r}{\sqrt{1 + 2rc + r^2}}\right) > 0, \qquad -1 \le c \le 1,$$

where $\sqrt{1 + 2rc + r^2} \ge 1 - r > r$, the global lower endpoint is $f(-1) = h \in (-1, 1)$. Thus μ_r vanishes on $(-1, h)$ but has positive mass in $(h, 1)$. The analytic density furnished by Lemma 2 would vanish identically on the connected interval $(-1, 1)$, a contradiction.

Case 2: $1/2 < r < 1$**.** Let $u = t - \pi$ and $v = \theta - \pi$. Expanding about this interior point of the torus gives

$$Y_r = h + \frac{2r(1 - 2r)}{1 - r}u^2 + 2r(1 - r)v^2 + O\big((|u| + |v|)^4\big). \tag{10}$$

The two quadratic coefficients are nonzero and have opposite signs. The Morse lemma supplies local smooth coordinates, with a bounded positive Jacobian, in which $Y_r - h = U^2 - V^2$. After a further invertible linear change, this is the product of the coordinates. In a fixed small square

the region where the absolute value of that product is at most ϵ has area at least $c\epsilon\log(1/\epsilon)$ for all sufficiently small $\epsilon > 0$. Since t and θ are uniformly distributed, it follows that

$$\mu_r([h-\epsilon, h+\epsilon]) \geq c'\epsilon\log(1/\epsilon).$$

Since $h \in (-1,1)$, this contradicts the $O(\epsilon)$ bound required by a density bounded near h.

Case 3: $r = 1/2$**.** Now $h = -1$ and direct expansion gives

$$Y_{1/2} = -1 + \tfrac{1}{2}(u^4 + v^2) + O(u^6 + u^2v^2 + v^4). \tag{11}$$

Together with (9), this implies $0 \leq Y_{1/2} + 1 \leq C(u^4 + v^2)$ locally. For a sufficiently small fixed $c > 0$, the rectangle

$$|u| \leq c\epsilon^{1/4}, \qquad |v| \leq c\epsilon^{1/2}$$

is mapped into $[-1, -1+\epsilon]$. It has probability at least $c'\epsilon^{3/4}$. This contradicts the $O(\epsilon)$ mass bound from the density bounded near -1 in Lemma 2. The three cases cover $0 < x = r^2 < 1$. □

5 A positive elliptic kernel for $x > 1$

Theorem 4. *For every real $x > 1$, the sequence $(A_n(x))_{n\geq 0}$ has a positive compactly supported Stieltjes representing measure.*

Fix $r = \sqrt{x} > 1$ and set

$$h = 1 + 8r^2 - 8r, \qquad H = 1 + 8r^2 + 8r, \qquad c_r = \frac{1}{2\pi^2 r}.$$

Write f_r for the density of the Legendre averaging law on $(-1,1)$. We will continue f_r to $\Omega = \mathbb{C} \setminus [1,\infty)$, express it as the Cauchy transform of a positive density, and convert its Legendre moments into power moments.

5.1 The density and its continuation

Throughout this section $\mathcal{K}$ uses a *parameter*, not a modulus:

$$\mathcal{K}(w) = \int_0^{\pi/2} (1 - w\sin^2\theta)^{-1/2}\, d\theta = R_F(0, 1-w, 1).$$

Here R_F is Carlson's symmetric integral, continued from

$$R_F(a,b,c) = \frac{1}{2}\int_0^\infty \frac{dt}{\sqrt{(t+a)(t+b)(t+c)}}.$$

Each factor uses its principal square root on the initial domain. These conventions agree with [6, Sections 19.16 and 19.25] after replacing its squared modulus by our parameter.

For $-1 < y < 1$, put

$$d = \sqrt{(y+1)/2}, \quad m = \frac{y-1}{4r}, \quad A = (1-d)^2 - m^2, \quad B = (1+d)^2 - m^2. \tag{12}$$

Conditional on $c = \cos t$, the second angular variable has an arcsine law. Its squared radicand factors as

$$16r^2(1 + 2rc + r^2) - (y - 1 - 4r^2 - 4rc)^2 = 16r^2\big[d^2 - (c-m)^2\big].$$

The conditional law has no atoms, since its amplitude is at least $4r(r-1) > 0$. Integrating against the arcsine density of c gives

$$f_r(y) = \frac{1}{4\pi^2 r} \int_{m-d}^{m+d} \frac{dc}{\sqrt{(1-c^2)(c-m+d)(m+d-c)}}.$$

Indeed, the roots of the conditional quadratic radicand are $m-d, m+d$, both strictly inside $(-1,1)$ when $r > 1$ and $-1 < y < 1$. Explicitly, $1+m-d = (1-d)(1-(d+1)/(2r)) > 0$ and $1-m-d = (1-d)(1+(d+1)/(2r)) > 0$. The four-root substitution

$$t^2 = \frac{(e_3-e_1)(c-e_2)}{(e_3-e_2)(c-e_1)}, \qquad (e_1,e_2,e_3,e_4) = (-1, m-d, m+d, 1),$$

reduces this integral to the standard complete elliptic integral. Since $(e_3-e_1)(e_4-e_2) = B$ and $(e_3-e_2)(e_4-e_1) = 4d$, it yields

$$f_r(y) = c_r R_F(0, A, B) = c_r B^{-1/2} \mathcal{K}(4d/B). \tag{13}$$

Lemma 5. *The germ in (13) extends to a single-valued holomorphic function on Ω.*

Proof. On the two-sheeted surface $d^2 = (z+1)/2$, the finite singularities of the composition in (13) can occur only above $-1, 1, h, H$. Indeed, $\mathcal{K}(p) = \pi\, {}_2F_1(1/2, 1/2; 1; p)/2$ continues along every path avoiding $p = 0, 1, \infty$, by the classical hypergeometric equation [6, 15.10.1]. Here $p = 4d/B$, and the preimages are $d = 0$, $A = 0$, or $B = 0$. The factorizations

$$A = (d-1)^2 \left(1 - \frac{(d+1)^2}{4r^2}\right), \qquad B = (d+1)^2 \left(1 - \frac{(d-1)^2}{4r^2}\right) \tag{14}$$

give exactly those projected points. Composing with the holomorphic map $d \mapsto p$ creates no new singularities, even at its critical points, and the prefactor $B^{-1/2}$ continues along every path that avoids the zeros of B. Thus the selected germ continues along every path in $\Omega \setminus \{-1\}$.

The point $z = -1$, where $d = 0$, is not a singularity of this germ. The change $c = m + d\cos\phi$ in the density integral gives, near $d = 0$,

$$f_r = \frac{1}{4\pi^2 r} \int_0^\pi \left[1 - \left(\frac{d^2-1}{2r} + d\cos\phi\right)^2\right]^{-1/2} d\phi.$$

At $d = 0$ the expression inside the inner square has modulus $1/(2r) < 1$. For sufficiently small complex d the integrand has a single holomorphic square-root branch uniformly in ϕ. Its integral is holomorphic in d, and $\phi \mapsto \pi - \phi$ shows it is even. It is therefore holomorphic in $z = 2d^2 - 1$ and agrees with the initial germ for real $d > 0$, so continuation around a small loop about -1 returns the same germ. Since Ω is simply connected and the fundamental group of $\Omega \setminus \{-1\}$ is generated by that loop, the monodromy theorem gives single-valuedness. The local integral then fills the puncture. □

5.2 The boundary values

In terms of the parameter, the classical connection formulas read

$$\mathcal{K}(z) = (1-z)^{-1/2} \mathcal{K}\big(z/(z-1)\big), \qquad z < 0, \tag{15}$$

$$\mathcal{K}(z) = z^{-1/2} \big[\mathcal{K}(1/z) + i\mathcal{K}(1-1/z)\big], \qquad z \text{ above } (1,\infty). \tag{16}$$

They hold as identities of the adjacent holomorphic germs, and the version of (16) below $(1,\infty)$ is its conjugate; see [6, 19.7.2–19.7.3] for the sign conventions on the two sides. The convergent logarithmic expansion [6, 19.12.1] gives

$$\mathcal{K}(1-\epsilon) = -\frac{\mathcal{K}(\epsilon)}{\pi}\log\epsilon + G(\epsilon), \tag{17}$$

where G is holomorphic near zero.

For real $y > 1$, retain the positive square root in d. The factorizations (14) show that A has a double zero at 1 and a simple zero at h, while B has a simple zero at H.

Lemma 6. *The upper boundary value of f_r is given by the following holomorphic germs adjacent to the indicated real intervals:*

$$f_r(z) = \frac{c_r}{\sqrt{B}}\big[\mathcal{K}(p) + 2i\mathcal{K}(1-p)\big], \qquad p = \frac{4d}{B}, \quad 1 < y < h, \tag{18}$$

$$f_r(z) = \frac{c_r}{\sqrt{4d}}\big[\mathcal{K}(q) + 3i\mathcal{K}(1-q)\big], \qquad q = \frac{B}{4d}, \quad h < y < H, \tag{19}$$

$$f_r(z) = \frac{c_r}{\sqrt{-A}}\big[-2\mathcal{K}(t) + 3i\mathcal{K}(1-t)\big], \qquad t = \frac{B}{A}, \quad H < y < \infty. \tag{20}$$

The parameters p, q, t respectively lie in $(0,1)$ on their displayed real intervals, and all displayed real prefactors use positive roots.

Proof. Near 1,

$$A = \frac{1-r^{-2}}{16}(z-1)^2 + O((z-1)^3), \qquad B = 4 + O(z-1).$$

Going around 1 through the upper half-plane makes A/B wind clockwise once about zero. Equation (17) adds exactly $2i\mathcal{K}(A/B)$, proving (18) near 1. On compact subintervals of $(1,h)$, both p and $1-p$ stay in $(0,1)$, so (18) holds on all of $(1,h)$ with no further change.

At h,

$$A'(h) = -\frac{(r-1)^2}{r(2r-1)} < 0, \qquad p'(h) = -A'(h)/B(h) > 0.$$

An upper semicircle about h thus sends p through the upper side of 1. Apply (16) to $\mathcal{K}(p)$ and (15) to $\mathcal{K}(1-p)$:

$$\mathcal{K}(1-p) = p^{-1/2}\mathcal{K}(1-1/p).$$

Substitution in (18) gives (19), including the coefficient 3. The square-root factors agree on these germs because $Bp = 4d$ and they agree before crossing. Again, compact subintervals of (h,H) keep $q, 1-q$ strictly between 0 and 1.

At H,

$$B'(H) = -\frac{(r+1)^2}{r(2r+1)} < 0, \qquad q'(H) = B'(H)/(4d(H)) < 0.$$

Hence q crosses zero through the lower half-plane and $1-q$ crosses 1 through the upper half-plane. Apply (15) to $\mathcal{K}(q)$ and (16) to $\mathcal{K}(1-q)$. With

$$\frac{q}{q-1} = t, \qquad \frac{1}{1-q} = 1-t, \qquad (4d)(1-q) = -A,$$

the real coefficient becomes $1-3 = -2$, giving (20). The parameters $t, 1-t$ remain in $(0,1)$ to the right of H, so (20) holds on all of (H,∞). By Lemma 5, these continuations do not depend on the path. □

Define $\rho_r(y) = \operatorname{Im} f_r(y + i0)/\pi$. Lemma 6 gives the density

$$\rho_r(y) = \frac{1}{2\pi^3 r} \begin{cases} 2R_F(0, 4d, B), & 1 < y < h, \\ 3R_F(0, 4d, B), & h < y < H, \\ 3R_F(0, -A, -B), & H < y < \infty. \end{cases} \tag{21}$$

It is positive, since $4d$ and B are positive on $(1, H)$, $-A$ and $-B$ are positive on (H, ∞), and $R_F(0, a, b) > 0$ for $a, b > 0$. Moreover, ρ_r has a finite right limit at 1, a finite jump at h, and at most a logarithmic singularity at H. At infinity, $-A, -B \sim y^2/(16r^2)$, so homogeneity gives $\rho_r(y) = O(1/y)$.

5.3 The Cauchy representation

Lemma 7. *For every $z \in \Omega$,*

$$f_r(z) = \int_1^\infty \frac{\rho_r(a)}{a - z}\, da. \tag{22}$$

Proof. The integral converges by the local logarithmic bounds and $\rho_r(a) = O(1/a)$. To apply Cauchy's theorem, we need a bound that is uniform around a large circle, including angles close to the cut.

Take $R_0 = 1000r^2$ and the connected upper exterior domain $D = \{z : \operatorname{Im} z > 0,\ |z| > R_0\}$. Choose $d = \sqrt{(z+1)/2}$ by continuation from the positive bank. The expansion

$$A = -\frac{z^2}{16r^2} + \left(\frac{1}{2} + \frac{1}{8r^2}\right) z + \frac{3}{2} - \frac{1}{16r^2} - 2\sqrt{(z+1)/2}$$

gives $|A + z^2/(16r^2)| \le 5R/4$, $R = |z| > R_0$, at every angle. Thus

$$\frac{R^2}{32r^2} \le |A| \le \frac{3R^2}{32r^2}, \qquad \frac{\sqrt{R}}{2} \le |d| \le \sqrt{R}.$$

Put $w = 1 - t = -4d/A$ and $A = -z^2(1 + e)/(16r^2)$. Then

$$\frac{64r^2}{3R^{3/2}} \le |w| \le \frac{128r^2}{R^{3/2}} < \frac{1}{2}, \qquad |e| \le \frac{1}{50}.$$

Take $0 < \arg z < \pi$, and use the principal logarithms of $1 + 1/z$ and $1 + e$, which are analytic here because $1/z$ and e are small. The logarithm of w continued from the positive bank is

$$\operatorname{Log} w = \log \frac{64r^2}{\sqrt{2}} - \frac{3}{2} \operatorname{Log} z + \frac{1}{2} \log(1 + 1/z) - \log(1 + e). \tag{23}$$

In particular,

$$|\operatorname{Im} \operatorname{Log} w| \le \frac{3\pi}{2} + \tfrac{1}{2} \arcsin(1/1000) + \arcsin(1/50) < 5, \qquad |\operatorname{Log} w| \le \tfrac{3}{2} \log R + 5.$$

The bounds $R^{-3/2} < |w| < 1$ control the real part. We keep this continued branch rather than the principal logarithm; its imaginary part approaches $-3\pi/2$ near the negative real axis. The square root of $-A$ on D is chosen positive at the positive bank; its modulus is $|A|^{1/2}$.

In (17), the convergent series [6, 19.12.1–19.12.3] is

$$G(w) = \sum_{j\geq 0} \left(\frac{(1/2)_j}{j!}\right)^2 [\psi(j+1) - \psi(j+1/2)]w^j.$$

Its coefficients satisfy $((1/2)_j/j!)^2 \leq 1$ and $0 < \psi(j+1) - \psi(j+1/2) \leq 2\log 2$. Hence for $|w| \leq 1/2$,

$$|\mathcal{K}(w)| \leq \pi/\sqrt{2}, \qquad |G(w)| \leq 4\log 2.$$

Equation (20) becomes

$$f_r(z) = \frac{c_r}{\sqrt{-A}} \left[\frac{2}{\pi}\mathcal{K}(w)\operatorname{Log} w - 2G(w) + 3i\mathcal{K}(w)\right]. \tag{24}$$

With these branches both sides are holomorphic on D, and they agree near the positive bank, so by the identity theorem they agree on all of D. On the whole upper semicircle this gives

$$|f_r(z)| \leq \frac{\sqrt{32}}{2\pi^2 R}\left[\sqrt{2}\left(\tfrac{3}{2}\log R + 5\right) + 8\log 2 + \frac{3\pi}{\sqrt{2}}\right].$$

Schwarz reflection gives the lower semicircle, and holomorphy supplies the negative-axis limit. Thus the estimate is uniform on the entire slit circle, and the large-circle Cauchy integral vanishes.

At 1, A/B has a double zero and (17) gives $f_r(z) = O(1 + |\log|z-1||)$. At h, $1-p$ has a simple zero in (18); at H, q has a simple zero in (19). The same logarithmic expansion therefore gives $f_r(z) = O(1 + |\log|z-a||)$ near each $a \in \{1, h, H\}$ on either side of the cut. An arc of radius ϵ contributes $O(\epsilon(1 + |\log\epsilon|))$, which tends to zero. Cauchy's theorem on the slit plane now gives (22), with $(f_+ - f_-)/(2\pi i) = \rho_r$ and denominator $a - z$. □

5.4 The positive remainder measure

Common's positive Froissart–Gribov theorem [1, CERN preprint, Section 4, Theorem 5] converts positive Q_n mixtures into Stieltjes moments. We give the integral calculation for the present density, whose spectral support reaches 1 and whose total spectral mass need not be finite.

Use the standard normalization

$$Q_n(a) = \frac{1}{2}\int_{-1}^{1} \frac{P_n(y)}{a-y}\,dy, \qquad a > 1.$$

Heine's integral is

$$Q_n(a) = \int_0^\infty \left(a + \sqrt{a^2-1}\cosh u\right)^{-n-1} du.$$

Set $t = (a + \sqrt{a^2-1}\cosh u)^{-1}$. Since $du = -dt/(t\sqrt{1-2at+t^2})$, this gives

$$Q_n(a) = \int_0^{a-\sqrt{a^2-1}} \frac{t^n}{\sqrt{1-2at+t^2}}\,dt. \tag{25}$$

In [6, 14.12.13–14.12.14], Olver's normalized function is $Q_n/n!$; multiplying both formulas by $n!$ yields this normalization.

Proof of Theorem 4. First take $n = 0$. All kernels in (22) and (25) are nonnegative on their real integration domains, so Tonelli's theorem gives

$$\int_{-1}^{1} f_r(y)\,dy = 2\int_{1}^{\infty} \rho_r(a)Q_0(a)\,da = \int_0^1 w_r(t)\,dt < \infty,$$

where w_r is defined below. The first integral is the probability $\mu_r([-1,1))$; the endpoints have zero mass by the conditional arcsine representation. For general n, put $b_n = \int_{-1}^{1} P_n(y)f_r(y)\,dy$. The bounds $|P_n(y)| \le 1$ and $0 \le t^n \le 1$ reduce absolute integrability to the case $n = 0$. Fubini's theorem now gives

$$b_n = 2\int_{1}^{\infty} \rho_r(a)Q_n(a)\,da = \int_0^1 t^n w_r(t)\,dt, \tag{26}$$

$$w_r(t) = 2\int_{1}^{(t+t^{-1})/2} \frac{\rho_r(a)}{\sqrt{1-2at+t^2}}\,da \ge 0, \qquad 0 < t < 1. \tag{27}$$

The domain in (27) follows from $t \le a - \sqrt{a^2-1}$ if and only if $1 \le a \le (t+t^{-1})/2$. Thus $w_r(t)\,dt$ is a finite positive measure on $[0,1]$ with moments b_n.

The contribution from $Y_r \ge 1$ already has the positive Laplace-mixture measure ν_+, supported on $[H - \sqrt{H^2-1}, H + \sqrt{H^2-1}]$. Adding the positive measure in (27) proves that $\nu_+ + w_r(t)\,dt$ is a positive compact Stieltjes measure with moments $A_n(x)$. □

Proof of Theorem 1. Theorem 3 excludes $0 < x < 1$ and Theorem 4 covers $x > 1$. Edgar's theorem [3] covers $x = 1$. At $x = 0$, $A_n(0) = 1$ is represented by the point mass at 1. □

6 Hankel certificates and fixed-size tests

We now give exact sign certificates on an interval of parameters. Define

$$H_{d,s}(x) = \det\big(A_{i+j+s}(x)\big)_{0\le i,j<d}, \qquad s \in \{0,1\}.$$

Proposition 8. *For every $0 < x \le 2/3$, at least one of $H_{4,0}(x), H_{8,0}(x), H_{9,1}(x), H_{10,0}(x)$ is strictly negative.*

Proof. Table 1 gives exact sign certificates. For each row, $Q(x) = H_{d,s}(x)/x^v$ is a polynomial of degree N.

Table 1: Intervals and polynomial degrees for the negative-minor certificates.

(d,s)	a	b	v	N
$(4,0)$	0	$9/100$	3	9
$(8,0)$	$1/20$	$101/500$	14	42
$(9,1)$	$201/1000$	$47/100$	18	63
$(10,0)$	$41/100$	$2/3$	23	67

Writing $Q(a+(b-a)t) = \sum_{j=0}^{N} c_j t^j$, its Bernstein coefficients are

$$\beta_k = \sum_{j=0}^{k} c_j \frac{\binom{k}{j}}{\binom{N}{j}}, \qquad 0 \le k \le N.$$

In each of the four rows, every β_k is a strictly negative rational number. The supplementary material records all coefficients; its verifier recomputes the determinants over $\mathbb{Q}[x]$, verifies their valuations and degrees, checks each rational sign, and reconstructs

$$Q(a+(b-a)t) = \sum_{k=0}^{N} \beta_k \binom{N}{k} t^k (1-t)^{N-k}.$$

The Bernstein basis is nonnegative and sums to one on $[0,1]$, proving $Q < 0$ throughout $[a,b]$. Hence $H_{d,s} < 0$ on that interval wherever $x > 0$. The four intervals overlap and cover $(0, 2/3]$. □

6.1 Finite tests near $x = 1$

No fixed finite set of Hankel minors detects every parameter in $(0,1)$. Indeed, Edgar's theorem [3, Theorem 1] provides a nonzero positive integrable moment density at $x = 1$. For any fixed d and $s \in \{0,1\}$, and any nonzero real vector v,

$$\sum_{i,j=0}^{d-1} v_i v_j A_{i+j+s}(1) = \int_0^\infty t^s \left(\sum_{i=0}^{d-1} v_i t^i \right)^2 d\nu(t) > 0.$$

A nonzero polynomial has finitely many zeros, whereas the absolutely continuous measure cannot be supported there. Thus each fixed Hankel matrix is positive definite at $x = 1$, and by continuity any finite collection of them, or of their principal submatrices, stays positive definite in a neighborhood of 1. At each $x \in (0,1)$, Theorem 3 gives a negative principal minor of the ordinary or shifted Hankel matrix, but it does not say which minor or bound its size.

6.2 Shifted Hankel matrices

Proposition 9. *For every fixed $x > 0$ and positive integer d, the matrix $(A_{N+i+j}(x))_{0 \le i,j < d}$ is positive definite for all sufficiently large integers N.*

Proof. Use (7) and the Laplace representation (2) to write $A_n(x) = a_n + b_n$, where $a_n = \int t^n \, d\nu_+(t)$ is the positive moment contribution from $Y \ge 1$, and $|b_n| \le 1$ is the contribution from $-1 \le Y < 1$. Put $r = \sqrt{x}$, $H = 1 + 8r^2 + 8r > 1$ and $R = H + \sqrt{H^2 - 1} > 1$. The law of Y has positive mass arbitrarily close to its maximum H, attained at $t = \theta = 0$ in (8). In the Laplace representation, varying its independent cosine angle gives ν_+ infinite support arbitrarily close to R. Choose $1 < c < R$ so that $\nu_+|_{[c,R]}$ has infinite support. Its degree-$(d-1)$ Gram matrix

$$G = \left(\int_{[c,R]} t^{i+j} \, d\nu_+(t) \right)_{0 \le i,j < d}$$

is strictly positive definite, since a nonzero polynomial cannot vanish on that support. Write $\lambda = \lambda_{\min}(G) > 0$. The matrix (a_{N+i+j}) dominates $c^N G$ in positive-semidefinite order, whereas (b_{N+i+j}) has operator norm at most d. Hence

$$\lambda_{\min}\big((A_{N+i+j}(x))_{0 \le i,j < d}\big) \ge c^N \lambda - d > 0$$

for all sufficiently large N. □

Thus fixed-size tests at increasingly large shifts become positive on both sides of the parameter threshold. The cutoff may depend on x and d.

Supplementary information

The supplementary material contains Python scripts and the exact rational certificates for Proposition 8, checks of the algebraic identities used in the proofs, and high-precision numerical checks of the elliptic continuation and the dispersion formula.

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